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Pharmacological Management of Mycobacterial Infections

Maria A. Pino, Ph.D
Department of Clinical Specialties
mpino@nyit.edu

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Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

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- Distinguish between the therapies used to manage latent vs. active tuberculosis.
- Describe RIPE treatment for tuberculosis. Explain the mechanisms of action, resistance, and significant adverse effects associated with these drugs.
- Review the second-line drugs used for tuberculosis, including aminoglycosides and fluoroquinolones.
- Describe the drugs used to manage nontuberculosis mycobacteria. Review the macrolide class of protein synthesis inhibitors
- Identify the drugs used in leprosy management, including dapsone and the highly teratogenic, thalidomide.

Source: Course Syllabus

Tuberculosis

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- High global mortality rates (underdeveloped areas, drug resistance)
- HIV comorbidity is a challenge
- At least 6 months required for effective treatment (multi-drug therapy)
- Goal: kill bacilli rapidly, prevent relapse, and disease transmission.
- Directly observed therapy (DOT) is necessary.
- Multidrug resistance: resistant to isoniazid or rifampin. Requires increased treatment and use of second-line drugs

Active Tuberculosis

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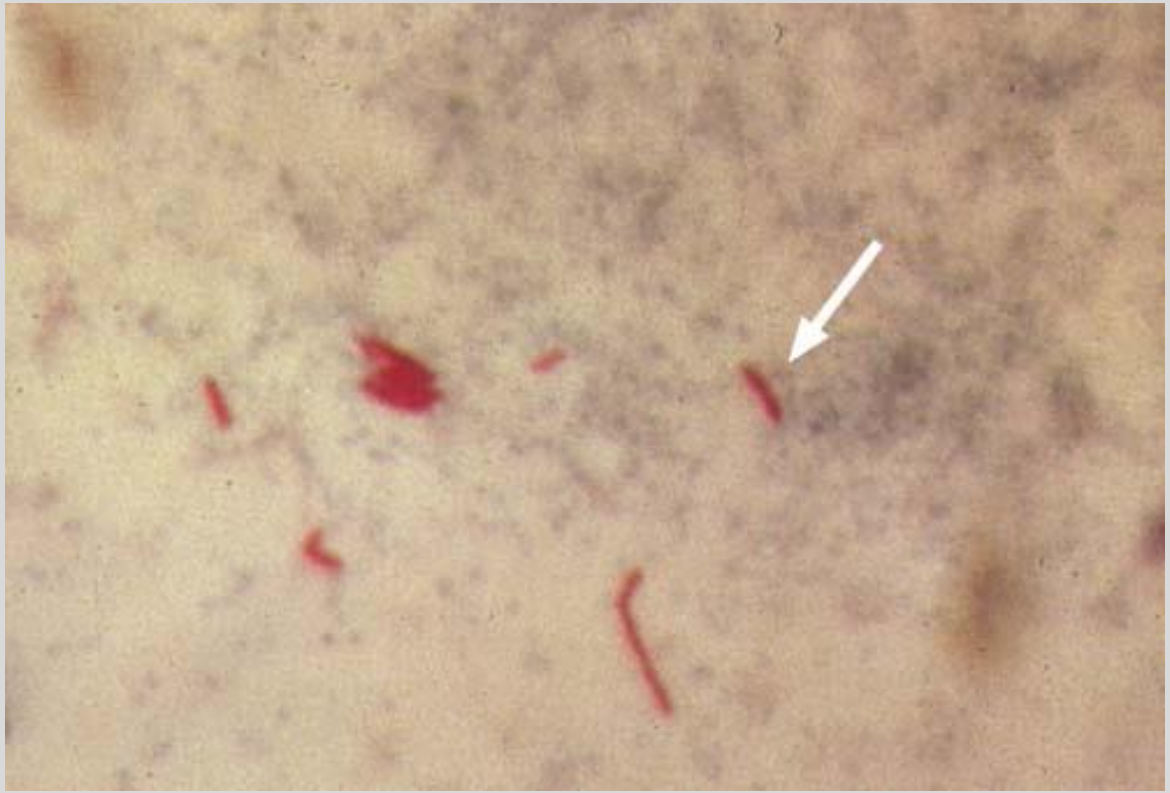
Presentation

- Cough
- Chest pain
- Blood in sputum
- Weakness or fatigue
- Weight loss
- Chills, fever, night sweats

Diagnosis

- Mantoux tuberculin skin test (PPD).
- Blood test (Quantiferon-TB Gold)
- Sputum smear
- Chest x-ray

Acid-Fast Bacilli



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Latent Tuberculosis

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- No symptoms
- Skin or blood test indicates TB; Normal chest x-ray and a negative sputum test
- Inactive TB in body, no spread of infection
- Requires drug prophylaxis
- TNF alpha inhibitors (etanercept, adalimumab) activate latent infections (Screen patient prior to initiation).

Active Treatment

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- **Isoniazid (INH), Ethambutol (EMB), Pyrazinamide (PZA), Rifampin (RIF). Treat with all four drugs for 2 months. Continue INH + RIF for 4 months (6 months total)**
- **Pregnancy: INH, RIF, EMB for 2 months. Continue INH + RIF twice weekly for 7 months. Not PZA or Streptomycin.**
- **Pediatrics: RIPE for 2 months, Continue INH + RIF for 4 months (6 months total). Non-severe TB can be treated for 4 months.**

Active Treatment, CDC 2025

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- 4 months of daily treatment with rifapentine, isoniazid, pyrazinamide and moxifloxacin.
- 8 weeks intensive, then 9 weeks continuous (but PZA is d/c)
- For those ages 12+, with no contraindications or drug resistance
- Vitamin B6 included for INH (manage neurological effects)

https://www.cdc.gov/tb/hcp/treatment/tuberculosis-disease.html?CDC_AAref_Val=https://www.cdc.gov/tb/topic/treatment/tbdisease.htm

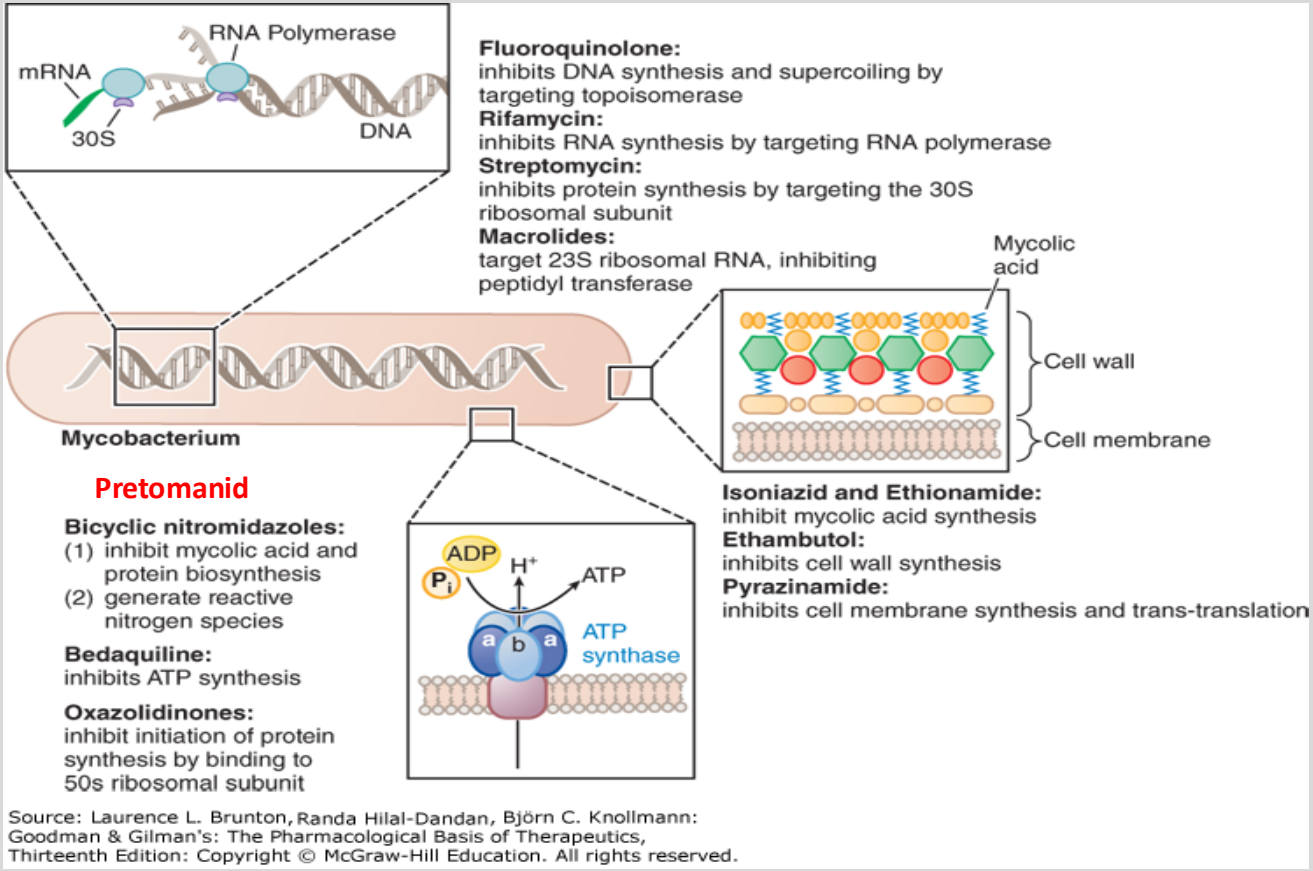
Latent Treatment

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- No HIV: INH + RIF daily for 3 months
- HIV, CD4 < 200: INH + rifapentine weekly for 3 months, or RIF daily for 4 months, or INH daily for 9 months.
- Pregnancy + HIV, CD4 < 350: RIF daily for 4 months.
- Pediatrics: RIF daily for 4 months

Drugs for Tuberculosis (MOA)

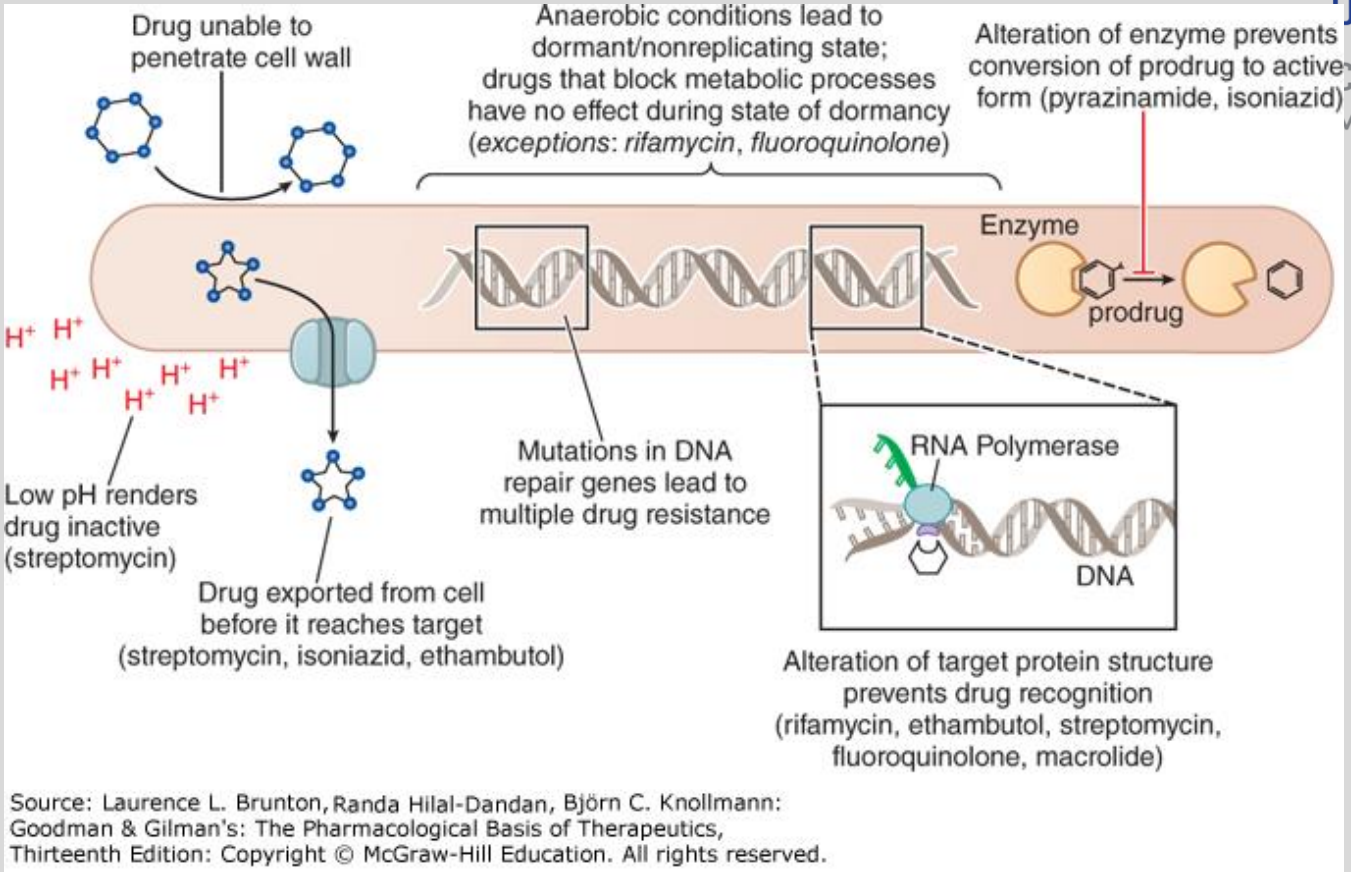
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Source: Laurence L. Brunton, Randa Hilal-Dandan, Björn C. Knollmann: Goodman & Gilman's: The Pharmacological Basis of Therapeutics, Thirteenth Edition: Copyright © McGraw-Hill Education. All rights reserved.

Drugs for Tuberculosis (Resistance)

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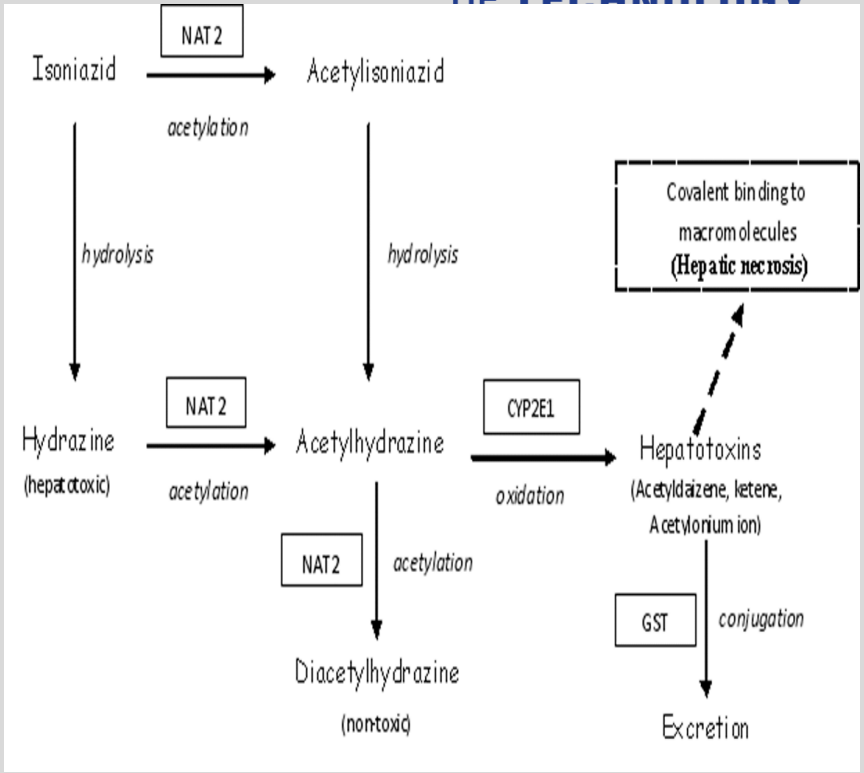
Isoniazid (INH)

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- Well-absorbed
- Structural similarity to pyridoxine
- Bactericidal; Inhibits synthesis of mycolic acid
- Activated by catalase-peroxidase (needs katG)
- Resistance results from mutated katG
- Acetylation (Drug-induced lupus)
- Adverse Effects: hepatotoxicity, especially in alcoholics, plus receiving PZA and RIF
- Peripheral neuritis (manage by B6; pyridoxine administration, 10-50 mg/day).
- Ethionamide similar in MOA (Second-line for TB)—activated by EthA

Isoniazid hepatotoxicity

- Acetylated toxic free radicals have been consistently implicated in the pathogenesis of INH hepatitis
- Drug-induced liver injury secondary to INH is more frequent in alcoholics or those taking rifampin and pyrazinamide.
- Fatal: requires liver transplantation.



Ethambutol (EMB)

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- Bacteriostatic
- Inhibits arabinosyl transferase encoded by the *emb* gene
- Treats TB and *Mycobacterium avium complex (MAC), M kansasii*
- Adverse effects: optic neuritis, impaired red-green color discrimination, hyperuricemia
- Resistance: mutations in *emb* gene

Pyrazinamide (PZA)

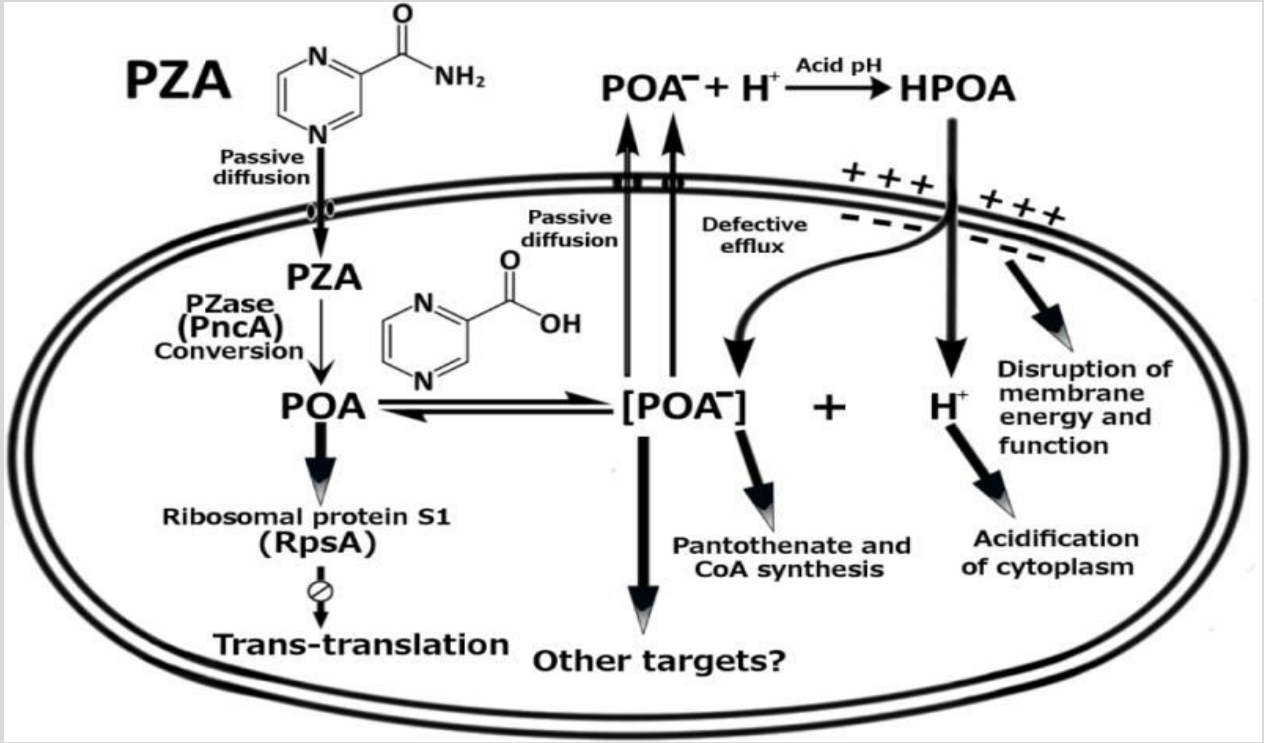
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- Bacteriostatic, but bactericidal on replicating bacteria
- Prodrug, needs Pzase for pyrazinoic acid conversion.
- Adverse effects: hepatotoxicity, hyperuricemia
- Resistance: mutations in genes involved in drug bioactivation by Pzase (PncA gene)

Pyrazinamide (PZA)

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https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4268777/

Rifampin (RIF)

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- Widely distributed into fluids (adverse effects)
- Inhibits DNA-dependent RNA polymerase
- Active against gram +, -, and acid-fast bacteria
- Prophylactic for *N. meningitidis, H influenzae*
- Adverse effects: Red secretions, hepatitis, CYP 3A4 inducer (Revs metabolism)
- Resistance: alteration in the DNA dependent RNA polymerase enzyme.
- "3 RRRs"

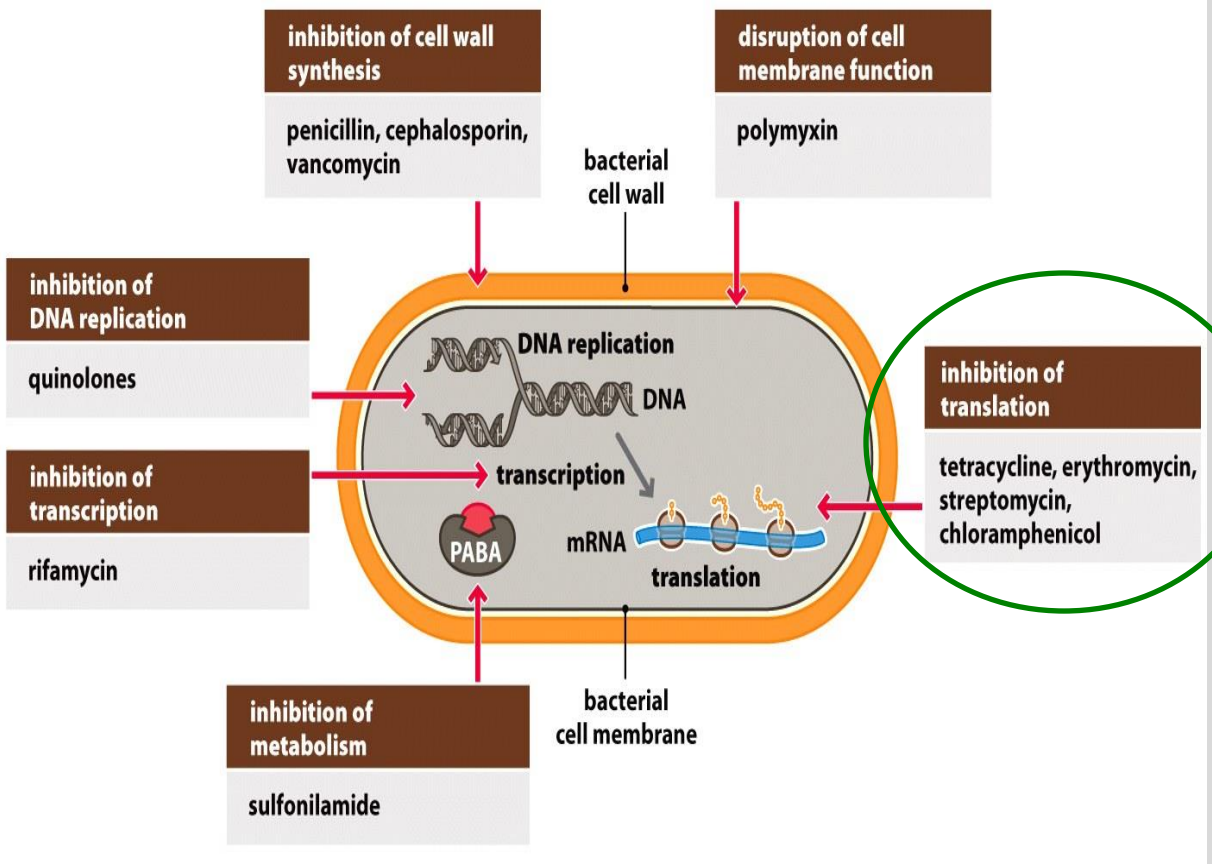


Second-Line Drugs and MDR strains

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- Aminoglycosides: capreomycin, amikacin, streptomycin (parenteral)
- Rifabutin
- Rifapentine
- Fluoroquinolones (Moxifloxacin)
- Azithromycin
- Ethionamide
- Cycloserine
- Bedaquiline: MDR strain
- Pretomanid: MDR strain
- Linezolid: MDR strain

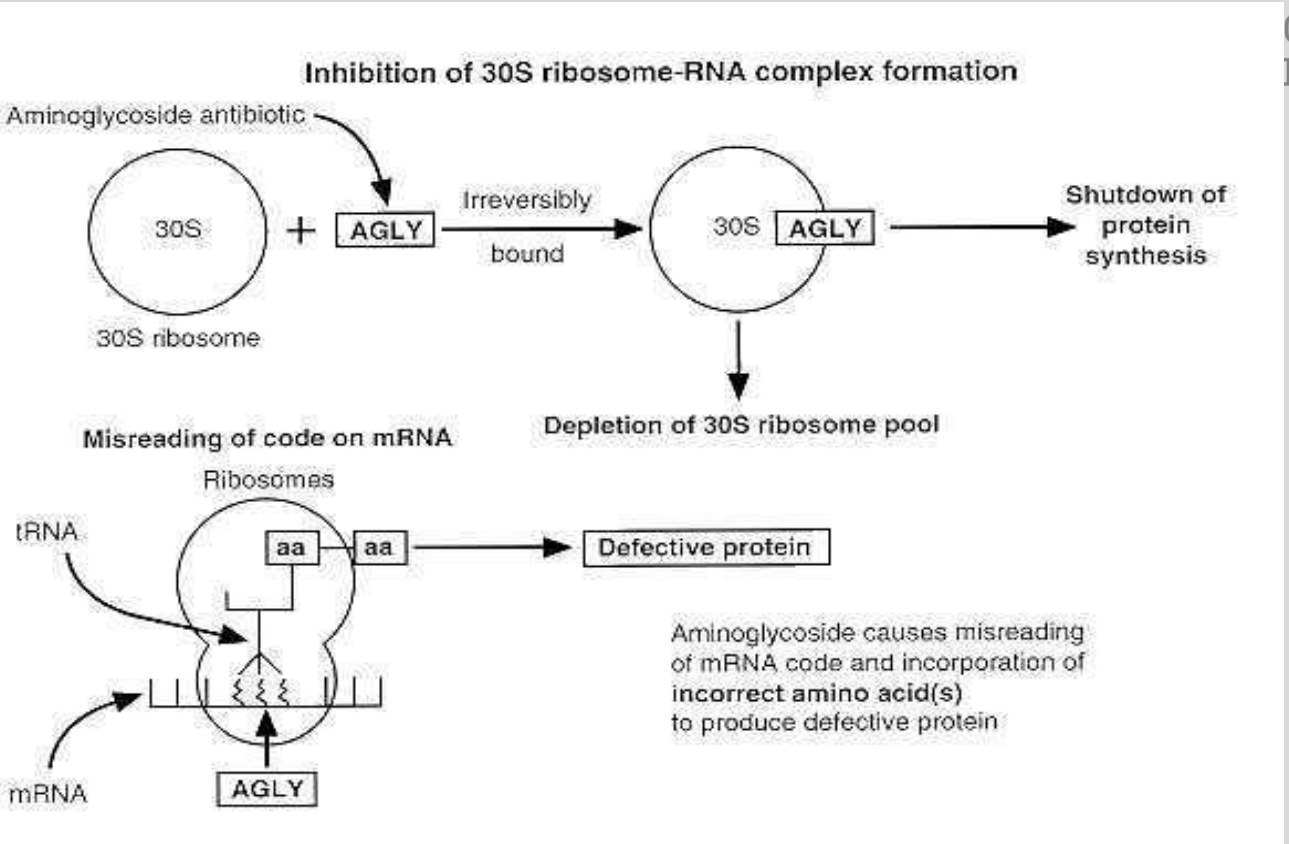
  "BPaL regimen"


inhibition of cell wall synthesis
penicillin, cephalosporin, vancomycin
bacterial cell wall
disruption of cell membrane function
polymyxin
inhibition of DNA replication
quinolones
DNA replication
DNA
inhibition of transcription
rifamycin
transcription
PABA
mRNA
translation
inhibition of translation
tetracycline, erythromycin, streptomycin, chloramphenicol
inhibition of metabolism
sulfonilamide
bacterial cell membrane

Aminoglycosides

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Aminoglycosides

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- Adverse effects: ototoxicity, nephrotoxicity, and neurotoxicity, teratogenic.
- Resistance: drug unable to bind ribosome, decreased uptake of drug, increased aminoglycoside modifying enzymes (acetyltransferases, adenyltransferases, phosphotransferases).

Aminoglycosides

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- **Treats: Gram-negative aerobic rods**
- **Post-Antibiotic Effect : Inhibits the growth of bacteria after removal of the drug. Immune system has time to remove bacteria.**
- **Concentration-dependent: Increased drug concentration compared to MIC = greater bactericidal action.**
- **Aminoglycosides and beta lactam antibiotics are synergistic and bactericidal (used for *Staph. Aureus*)**

Aminoglycoside adverse effects

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Neuromuscular Blockade

- Decreases acetylcholine release from the pre-synaptic neuron
- Avoid in anesthesia with neuromuscular blockade, myasthenia, and Parkinsonism.

Ototoxicity

- Toxic to both hearing and balance.
- Overlap toxicity with another ototoxic drug (loop diuretics, cisplatin)
- Cranial nerve 8 damage (pregnancy)

Source: Course Syllabus

Aminoglycoside adverse effects

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Nephrotoxicity

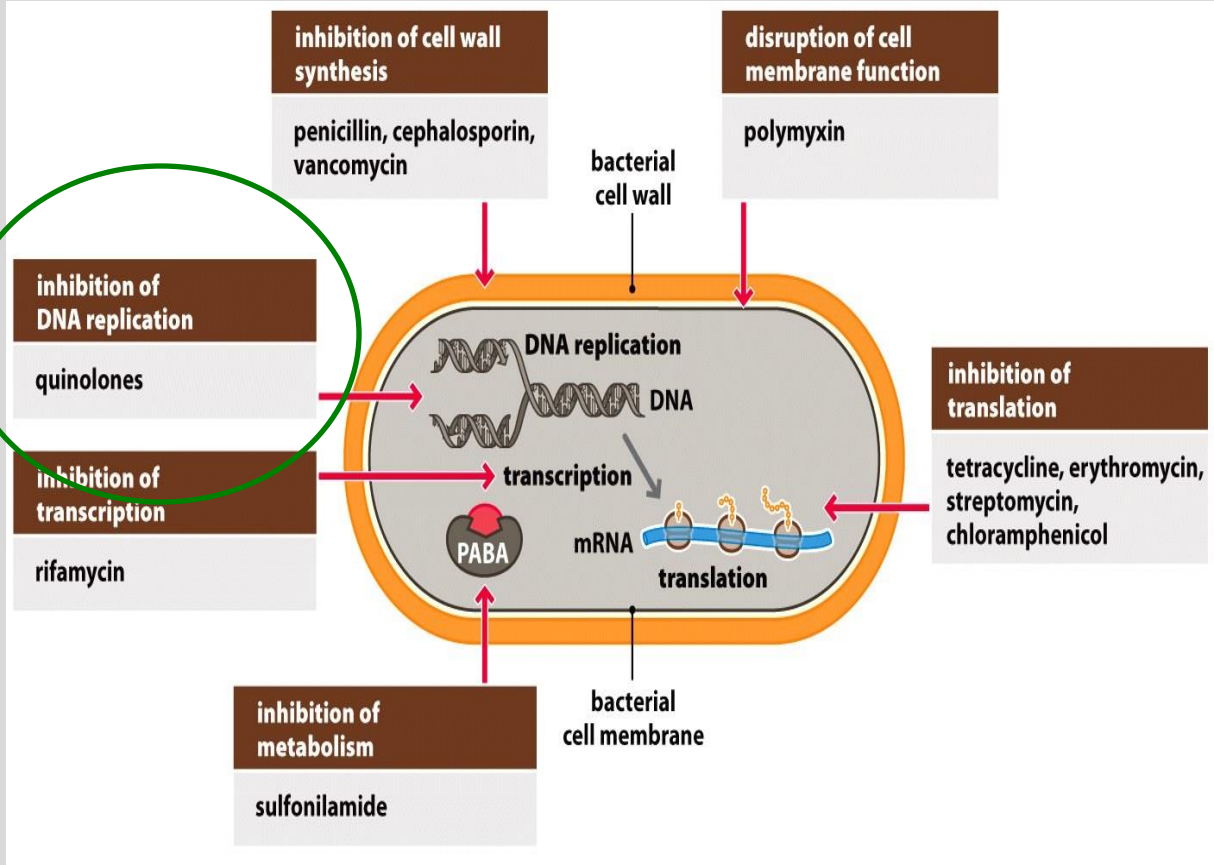
- Acute Tubular Necrosis (ATN). Loss of urine concentrating ability. Non-oliguric renal failure.
- Caution in patients taking other nephrotoxic drugs*

Source: Course Syllabus

Rifapentine and Rifabutin

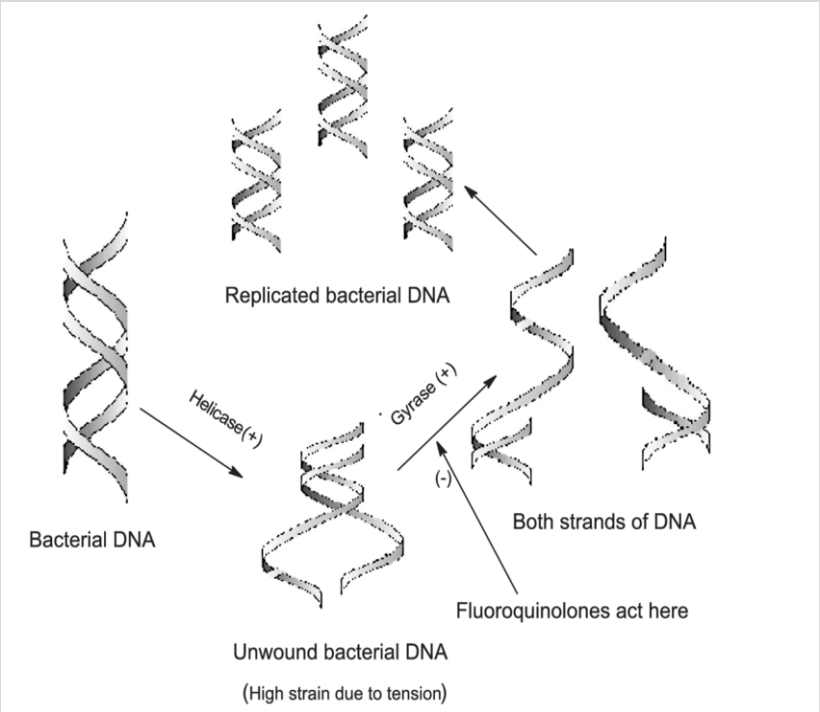
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- **Rifapentine: treats MAC, *M. tuberculosis.* Potent CYP induction and adverse effects as rifampin. Long half-life (allows for once-per-week dosing).**
- **Rifabutin: treats MAC, *M. tuberculosis.* Less CYP induction than rifampin. Replaces rifampin for HIV patients treated with HAART therapy to reduce drug interactions. Similar hepatotoxicity to rifampin. Reduce dose if CrCl is <30 mL/min.**


inhibition of cell wall synthesis
penicillin, cephalosporin, vancomycin
disruption of cell membrane function
polymyxin
bacterial cell wall
inhibition of DNA replication
quinolones
DNA replication
DNA
inhibition of translation
tetracycline, erythromycin, streptomycin, chloramphenicol
transcription
rifamycin
PABA
mRNA
translation
inhibition of metabolism
sulfonilamide
bacterial cell membrane

Fluoroquinolones “floxacins”

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- Use: Gram – rods of urinary and GI tract, *Pseudomonas, Neisseria*, *Shigella, E coli, Proteus, Klebsiella, Anthrax* and some gram + infections. Bactericidal
- Block DNA gyrase and topoisomerase IV.
- Inhibits the negative supercoiled configuration of the bacterial chromosome during and following DNA replication.

https://www.researchgate.net/figure/Schematic-representation-of-mechanism-of-action-of-fluoroquinolones_fig1_40849381

Fluoroquinolones

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- **Adverse Effects:** Increased QT interval
- Damage growing cartilage
- Category C (pregnancy)
- Tendon rupture
- Photosensitivity
- **Drug interactions**: CYP 1A2, di, trivalent cations
- **Resistance:** Reduced uptake of antibiotic, alteration of DNA gyrase

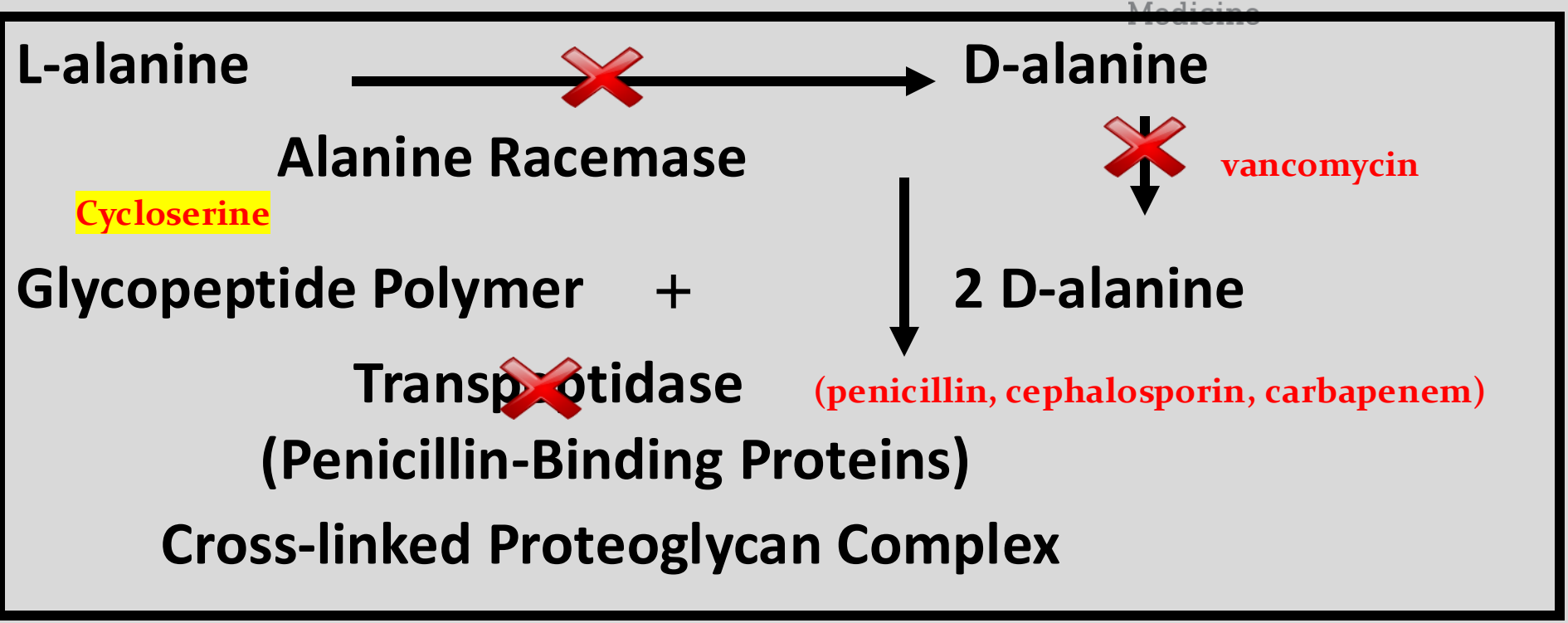
Moxifloxacin, Levofloxacin

- "Respiratory" fluoroquinolones
- *Legionella, Mycoplasma, Chlamydia, Klebsiella, Mycobacterium avium-intracellulare, M. tuberculosis, H. influenzae, S. pneumoniae.*

Source: Course Syllabus

Cell Wall Synthesis

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Cycloserine

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- MDR TB
- Inhibition of GABA transaminase. Partial NMDA agonist
- Resulting neurological symptoms limits use.

Linezolid

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- Binds 50S subunit
- Gram + activity
- Use: penicillin resistant *S. pneumoniae* and methicillin and vancomycin-resistant *Staphylococci* and vancomycin-resistant *Enterococci.* Multidrug resistant strains of TB
- Good GI absorption. 100% bioavailability.
- IV and oral dosing on same schedule.
- Adverse effects: Weak MAO inhibitor, caution with serotonin-increasing drugs. Bone marrow suppression.
- Resistance: mutation in 23S rRNA gene

Bedaquiline

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- Inhibits ATP synthase in mycobacteria
- Active against replicating and non replicating bacilli
- Metabolized by CYP P450 system (drug interactions)
- Long $t_{1/2}$ life
- Adverse effects: hepatotoxicity and cardiac toxicity (Increased QT interval-Blackbox warning)

Pretomanid

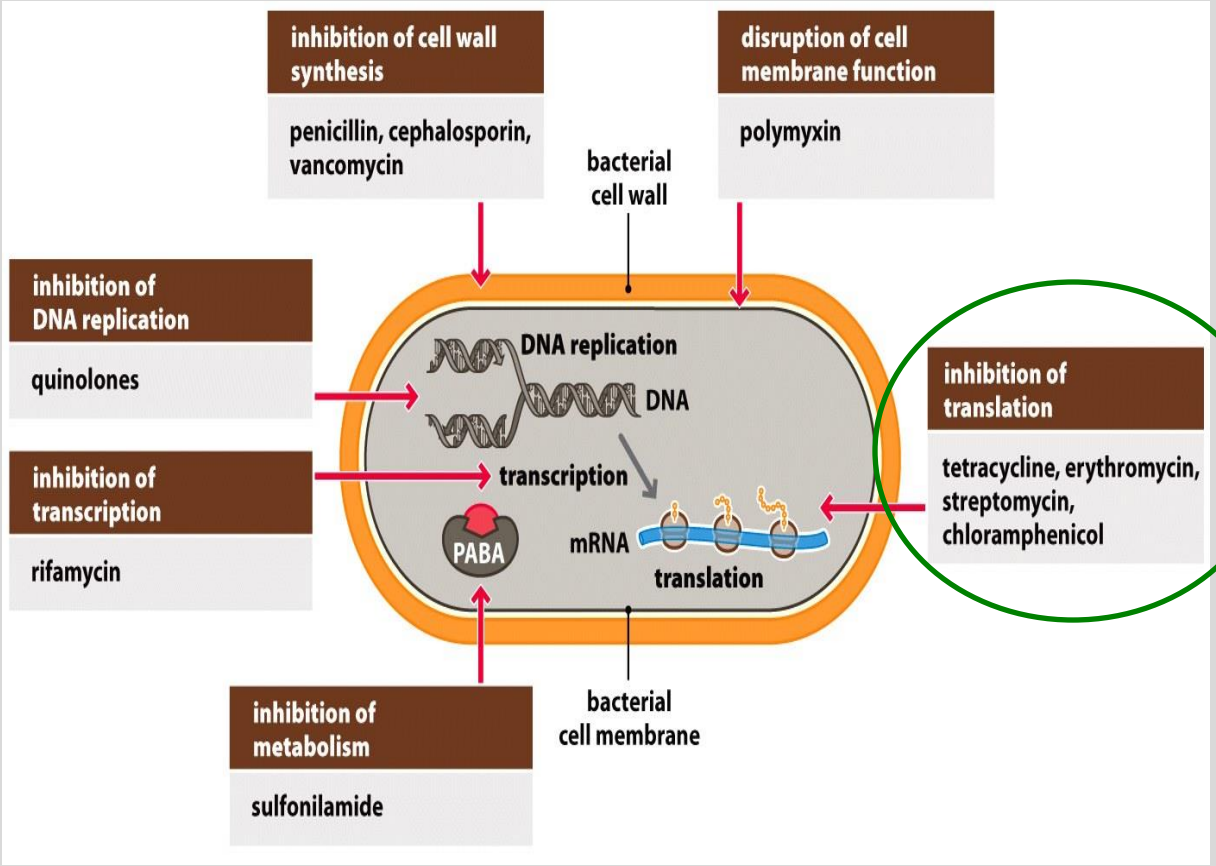
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- Nitroimidazole (prodrug), activated by the Ddn enzyme. ”think metronidazole”
- Increases NO to disrupt mycolic cell wall.
- Combined with bedaquiline and linezolid- **BPaL**
- Treat patient for ~ 6 months
- Indicated for pulmonary extensively drug resistant (XDR), or treatment-intolerant or nonresponsive multidrug-resistant (MDR)
- Long $t_{1/2}$
- Adverse effects: neuropathy, increased QT interval, hepatotoxicity

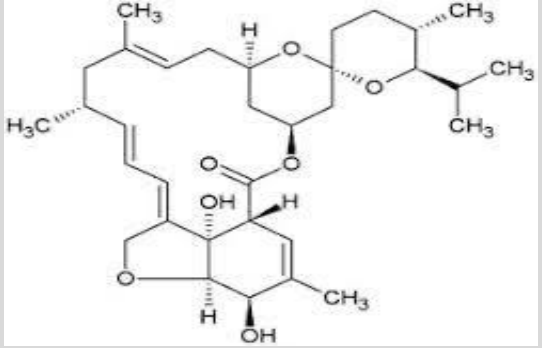
Mycobacterium avium complex (MAC)

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- **Presentation: fever, night sweats, weight loss, increase alkaline phosphate, anemia.**
- **Azithromycin or clarithromycin, plus ethambutol. Rifabutin can be added as a 3rd agent.**
- **Prevents MAC in HIV patients with a $CD_4 < 50$.**


inhibition of cell wall synthesis
penicillin, cephalosporin, vancomycin
bacterial cell wall
disruption of cell membrane function
polymyxin
inhibition of DNA replication
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rifamycin
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PABA
mRNA
translation
inhibition of translation
tetracycline, erythromycin, streptomycin, chloramphenicol
inhibition of metabolism
sulfonilamide
bacterial cell membrane

Macrolides (Azithromycin, Clarithromycin, Erythromycin)



- Bacteriostatic Proteins Synthesis Inhibitors
- Macrolide because of chemical structure (Macrocyclic lactone)
- Binds to 50S ribosome; Inhibits translocation
- Eythromycin is a CYP 3A4 inhibitor
- Macrolides safe in pregnant and pediatric patients
- Use: *Chlamydia, Mycoplasma, Legionella, Corynebacterium Diphtheria*, pertussis, syphilis.
- Resistance: efflux of drug, methylation of the 23S ribosomal binding site

Macrolides

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Clarithromycin:

- More active against *Strep.* and *Staph*. *H pylori, Moraxella cattarhalis, Legionella, M. pneumoniae, Borrelia*
- Well absorbed, less GI effects than erythromycin.
- Twice daily dosing
- CYP inhibition

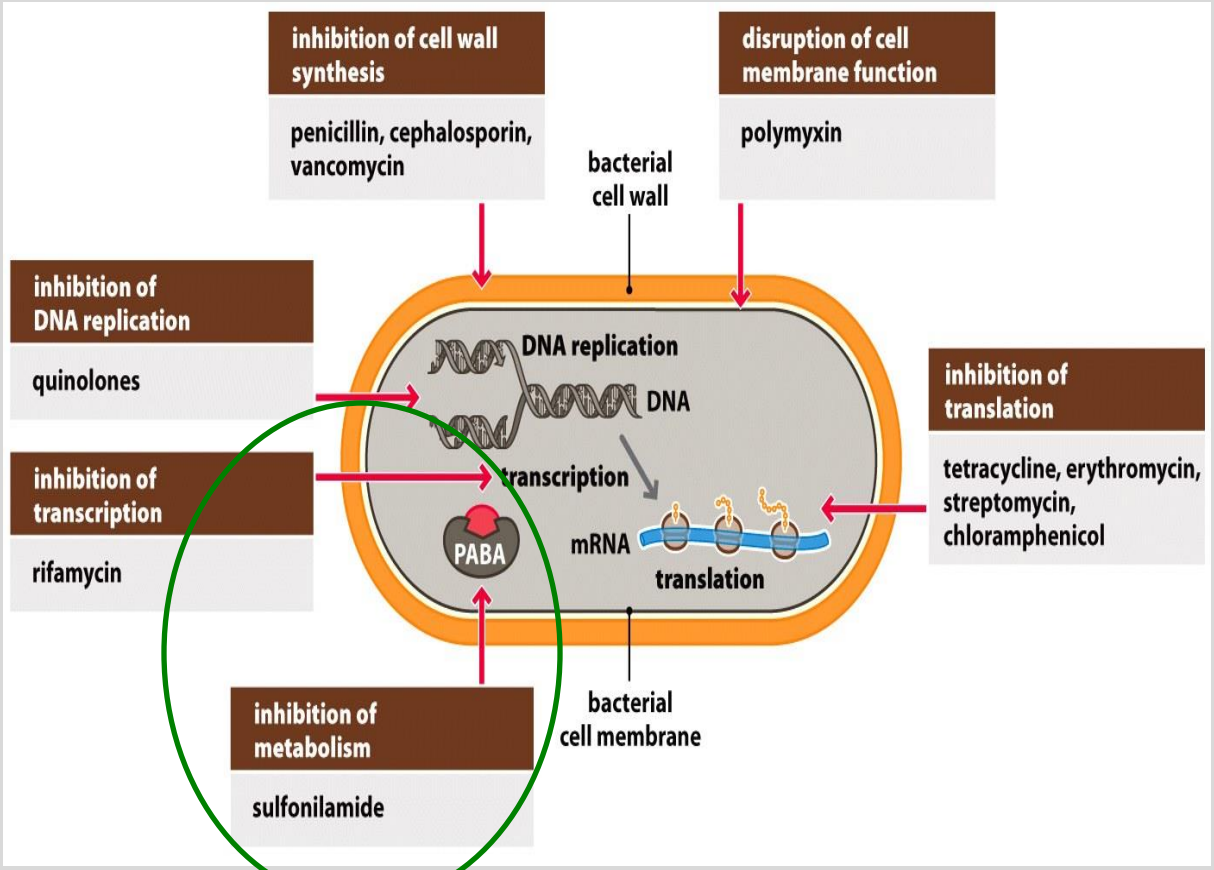
Azithromycin (Zithromax®):

- Increased activity against *H. Flu, Salmonella/Camphylobacter, Chlamydia*
- Long $t_{1/2}$, Once a day dosing
- No CYP inhibition

Hansen disease

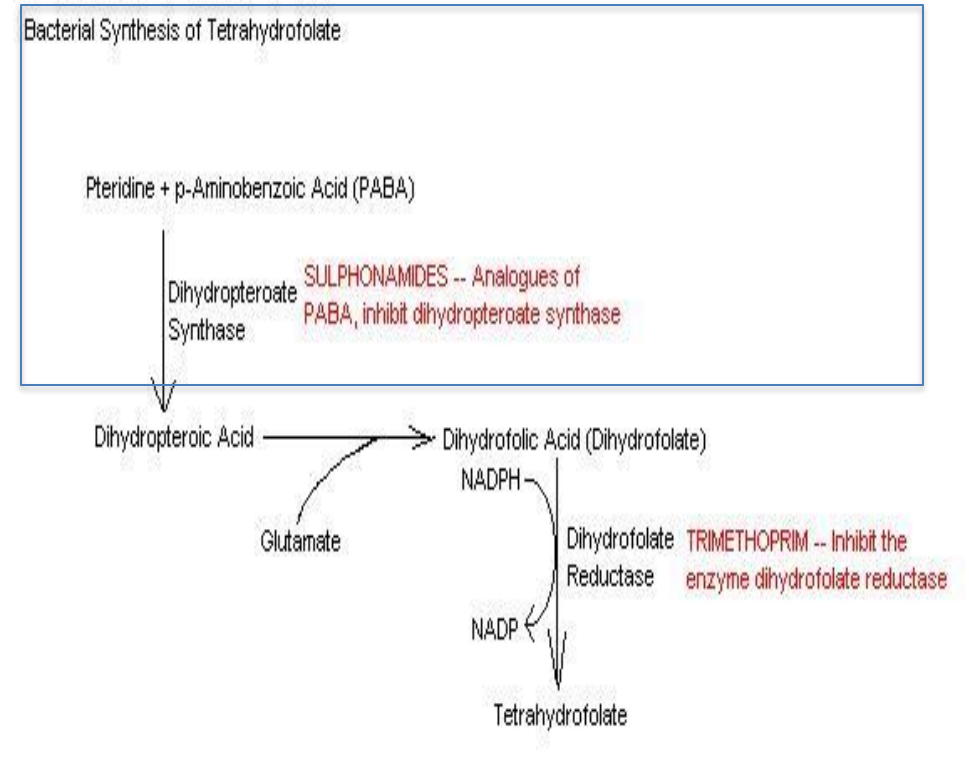
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- *Mycobacterium leprae*
- Leprostatic drugs: Dapsone, Rifampin, Clofazimine, or Thalidomide.
- Dapsone used for *Pneumocystis* prophylaxis in HIV patients. Topical use for severe acne.
- Thalidomide for erythema nodosum leprosum (TNF-alpha suppression).


inhibition of cell wall synthesis
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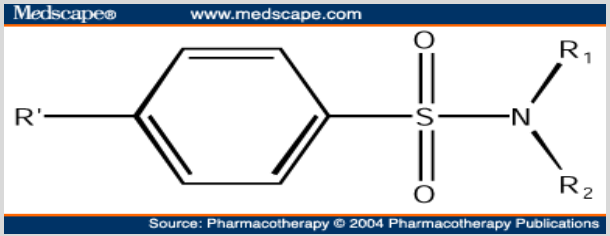
Dapsone MOA

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Sulfonamides:

- Synthetic antibiotics containing sulfur.
- Bacteriostatic
- Analogs of PABA



Dapsone

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- **Resistance:** Alteration of dihydropteroate synthase, decreased antibiotic uptake, increased PABA production.
- **Adverse effects: (similar to sulfonamides):** crystalluria, interstitial nephritis, G6PD deficiency, kernicterus (neonate), rash (erythema multiforme), photosensitivity, hypersensitivity, cross allergy with certain diuretics, hypoglycemic drugs

Clofazimine

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- Lipophilic, long duration of bactericidal action
- MOA: anti-inflammatory action (suppresses T-lymphocytes); Binds to guanine of DNA.
- Adverse effects: Discolors skin and conjunctiva, GI upset, increase QT interval. Not in pregnancy.
- Not for commercial use.
- Physician required to register as an investigator with the NHDP. When approved, the medication is dispensed from NHDP to the prescriber (National Hansen's Disease Program)

Thalidomide

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- Reduces the levels of TNFα, decreases angiogenesis. Treatment of multiple myeloma, leprosy
- Fetal effect: Phocomelia (short or absent long bones of the limbs).
- Drug is racemic; (R) enantiomer is effective against morning sickness but the (S) is teratogenic
- 1st trimester danger



Summary Slide

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- Describe RIPE treatment in tuberculosis. Explain the mechanisms of action, resistance, and significant adverse effects associated with these drugs.
- Identify the drugs used to manage nontuberculosis mycobacteria and leprosy.

Core References:

- Katzung & Trevor Basic and Clinical Pharmacology, 16e, 2024; Chapter 47, Antimycobacterial Drugs.
- Goodman and Gilman's The Pharmacological Basis of Therapeutics, 14e, 2023; Chapter 65, Chemotherapy of Tuberculosis, *Mycobacterium avium complex* Disease, and Leprosy
- Pharmacological Management of Mycobacterial Infections Summary Guide.
- Scholar Bricks: See CPG

Lecture Feedback Form:

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https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R